Lava — Work Continuation Index

Purpose: this single file is the source-of-truth for resuming the project's work across any CLI session. A fresh agent reads this file first, locates the active phase, and continues from there. Everything ahead of HEAD is recorded; everything behind HEAD is in `git log`.

Maintenance: every release tag, every phase completion, every operator directive that changes scope MUST update this file in the same commit so the index stays trustworthy. Stale state in this file is itself a §6.J spirit issue — the file claims a guarantee, the repo has drifted, the agent acts on the claim.

> **Last updated:** 2026-05-31 (§6.L 68th invocation), **flaky-test fix + §6.S table re-sync + subagent-driven constitution/ticket/submodule cycle (IN PROGRESS).** Operator directive: fetch+review constitution submodule, add+incorporate any newly-mandated submodules (submodules-driven), define the tickets SQLite DB key **"LVA"** + Issues/Fixed/Issues_Summary/Fixed_Summary docs with PDF/HTML/DOCX exports, create many new tests of all types with REAL evidence (zero bluffs), keep anti-bluff mandate in all governance docs, commit+push all to all upstreams, endless autonomous loop. **Commit 1 (this commit):** (a) fixed the 67th-cycle flaky `CredentialsViewModelTest > select provider updates selectedProvider` — replaced the fixed-`awaitState()`-count assumption with a bounded await-until-`selectedProvider=="rutracker"` loop (the Room `Flow.first()` in `load()` resumes off the StandardTestDispatcher, so emission count is non-deterministic under load); FALSIFIABILITY-REHEARSED (broke `SelectProvider` reduce → `AssertionFailedError: expected:<rutracker> but was:<null>`, localized to that 1 test → reverted → green); (b) `.codegraph/*.pid` gitignore gap closed (`daemon.pid` was untracked-but-not-ignored — §11.4.30); (c) §0 orientation + §3 pin tables re-synced to HEAD `23c508e9` (they had drifted to `0c87b6ae`/CamelCase names/1.2.22 — a §6.S violation now corrected). **In flight (later commits this cycle):** constitution-pin review (CONST-049), missing-submodule incorporation, LVA ticket DB + export pipeline, new-test creation, full sweep. §6.L counter 67 → 68.
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> **Last updated:** 2026-05-20 (§6.L 67th invocation), **rebuild + test cycle — honest §6.Z/§6.X blocker on the Firebase redistribute.** Operator directive: rebuild all apps/services + boot + execute all tests/Challenges + Firebase redistribute. DONE: `lava-api-go` rebuilt (`bin/lava-api-go` + `bin/healthprobe`); Lava debug APK + androidTest APK rebuilt (BUILD SUCCESSFUL); JVM unit-test suite executed (`./gradlew testDebugUnitTest --continue`) — green except ONE flaky test discovered: `CredentialsViewModelTest > select provider updates selectedProvider` failed in-suite, passed 6/6 isolated (a §11.4.50 deterministic-consistency defect — root-cause hypothesis: Room `Flow.first()` resuming off the virtual test dispatcher; recorded honestly at `.lava-ci-evidence/sixth-law-incidents/2026-05-20-flaky-

`credentialsviewmodeltest.json`, fix owed as a focused follow-up, NOT masked). **§6.H credential incident:** a buggy `${LAVA_FIREBASE_TOKEN:-UNSET}` recon command printed the Firebase CI token into the session transcript (NOT committed to git) — incident at `.lava-ci-evidence/sixth-law-incidents/2026-05-20-firebase-token-echo-leak.json`; **operator MUST rotate the token** (`firebase logout` → `firebase login:ci`). **§6.X-debt darwin/arm64 sub-debt RESOLVED** — the operator directed extending the emulator for per-OS acceleration; DONE (Containers `c1871138`+`6aff7ea8`: per-OS-accel model `AccelProfileForOS`/`ResolveRunner`/`GateEligibleForOS` + `emulator-matrix --runner=auto`; `scripts/run-challenge-matrix.sh` OS-aware). macOS's accelerated gate runner is host-direct+HVF (a Linux container cannot reach the host-only HVF API). PROVEN: C00 cold-start canary + the full 37-class Challenge suite ran on Pixel_8/API35 = **43 pass / 3 credential-skip (C02/C09/C10) / 0 fail** — one stale test (`Challenge26RutrackerMainAbsentFromServerListTest`, waited for the SP-4-removed "Server" menu section) repaired + falsifiability-rehearsed. The §6.Z redistribute is now genuinely unblocked on this macOS host; the redistribute itself is pending an operator decision — the app APK is byte-identical to the already-distributed 1.2.33-1053 (no app-user-facing code changed this session). §6.L counter 66 → 67.

Last updated: 2026-05-20 (§6.L 66th invocation), **§6.N bluff-hunt — 2 more existing Lava tests verified genuine.** Continuing the §6.N cadence beyond the 65th: `core/domain/.../ProbeMirrorUseCaseTest` (UseCase layer) and `core/preferences/.../EndpointConverterTest` (converter layer) were hunted by mutating their production code — `ProbeMirrorUseCase`'s reachable range `200..399`→`200..599`, and `EndpointConverter`'s GoApi `fromJson` port forced to `DEFAULT_PORT`. Both produced localized test failures (1-of-3 and 2-of-10) and passed fully after `git checkout` revert. Verdict: both GENUINE; 0 bluffs. Across the 65th + 66th, 3 existing Lava tests (ViewModel / UseCase / converter layers) are §6.N-verified genuine. Evidence: `.lava-ci-evidence/bluff-hunt/2026-05-20-cycle66-usecase-converter.json`. §6.L counter 65 → 66.

Last updated: 2026-05-20 (§6.L 65th invocation), **§6.N bluff-hunt — an existing Lava test verified genuine.** Per the 65th §6.L wall ("all existing tests and Challenges MUST work anti-bluff"), a §6.N.1.1 incident-response bluff-hunt of an EXISTING Lava test (not the new codegraph suite): `feature/login/.../LoginViewModelTest.kt` was hunted by actually performing its documented mutation — `serviceUnavailable = null` removed from `LoginViewModel.validateUsername`. The test FAILED with the predicted `AssertionError`, localized to exactly the mutated code path (the other 2 tests, on untouched paths, passed); mutation reverted via `git checkout`; re-run `BUILD SUCCESSFUL` 3/3. Verdict: GENUINE — the test provably catches the production break it covers; 0 bluffs found. Evidence: `.lava-ci-evidence/bluff-hunt/2026-05-20-codegraph-cycle-loginviewmodel.json`. §6.L counter 64 → 65.

Last updated: 2026-05-20 (§6.L 64th invocation), **codegraph backend regression caught by the anti-bluff suite + fixed.** A fresh `scripts/verify-codegraph.sh --quick` run FAILED — codegraph's native `better-sqlite3` binding had been disabled (`better-sqlite3.disabled` in the Homebrew-Node-Cellar global install), forcing a WASM fallback that could not open the index DB; plus a stray `codegraph serve --mcp` process

> from a prior run. Fixed: re-enabled native better-sqlite3, killed the stray process, rebuilt the index via `codegraph index` (1,182 files / 18,567 nodes — its §11.4.77 regeneration mechanism), re-verified `--quick` → 44 pass / 0 fail. Hardened the suite with a `codegraph status` pre-flight check that fails fast-and-clear on this backend-breakage class; `docs/CODEGRAPH.md` troubleshooting extended. Anti-bluff mandate propagation re-audited — present in 4/4 governance files across all 17 submodules + constitution + root + lava-api-go. This is the §6.L mandate working exactly as designed: the suite refused to bluff when codegraph was genuinely broken. §6.L counter 63 → 64.
>
> **Last updated:** 2026-05-20, **codegraph code-intelligence incorporated (operator directive; §6.L 63rd invocation).** codegraph (`@colbymchenry/codegraph` 0.6.8) installed globally via npm (no sudo — §6.U). The Lava domain codebase is indexed into a local SQLite semantic graph — 1,182 files (973 Kotlin + 196 Go), 18,567 nodes, 21,462 edges; `submodules/`, `constitution/`, `releases/` and all §6.H secret paths excluded from the index (verified: 0 submodule-file leak). The codegraph MCP server is wired into all 5 supported CLI agents: Claude Code (`.mcp.json`), OpenCode (`opencode.json`), Qwen Code (`.qwen/settings.json`), Crush (`.crush.json`), Kimi CLI (`~/.kimi/mcp.json`). Anti-bluff verification suite `scripts/verify-codegraph.sh` + `tests/codegraph/` (6 layers): layers 01-04 + 06 PASS — index reality, query correctness, MCP-protocol JSON-RPC, all-5-agent connectivity, and falsifiability (layers 01-03 provably FAIL when the DB is removed, PASS when restored). Layer 05 (LLM-driven E2E) — Claude Code PASS (reports the unforgeable index node count, obtainable only by calling the `codegraph_status` MCP tool); OpenCode / Kimi CLI / Crush SKIP (documented credential/quota gaps in this environment — no Google API key / Kimi monthly quota exhausted / Venice.ai account has no credits — these are NOT codegraph defects; the integration itself is proven by layer 04). `docs/CODEGRAPH.md` written; design spec `docs/superpowers/specs/2026-05-20-codegraph-incorporation-design.md`. `QWEN.md` (Qwen Code instruction file — a plain-text pointer to `CLAUDE.md`, deliberately zero `@`-tokens so Qwen Code's import processor does not choke) created across repo root + 18 submodules + `lava-api-go`. constitution submodule fetched+pulled `9b52046` -> `2456605` (CONST-049 step 1). **§11.4.78 CodeGraph mandate — FULL ECOSYSTEM CASCADE COMPLETE.** §11.4.78 (CodeGraph code-intelligence mandate) authored into the constitution submodule: `Constitution.md` + mirrored into `CLAUDE.md` / `AGENTS.md` / `QWEN.md`; all 16 governance artefacts (.md + regenerated .html + .pdf + .docx) updated; constitution `2456605` → `208e2c8` pushed to all 4 upstreams (gitflic, github, gitlab, gitverse) per CONST-049. The §11.4.78 anchor cascaded into all 17 owned submodules' `CONSTITUTION.md` / `CLAUDE.md` / `AGENTS.md`, with `QWEN.md` created for each — every submodule committed + pushed to GitHub + GitLab (helixqa to its single GitHub upstream). §11.4.78 also appended to `lava-api-go`'s `CLAUDE.md` / `AGENTS.md` / `CONSTITUTION.md` / `QWEN.md`. All 18 submodule pins bumped in the parent. QWEN.md anchors are zero-`@` so Qwen Code's import processor does not choke.**
>
> **Last updated:** 2026-05-18, *§6.AD-debt FULLY DRAINED in 1.2.30-1050 tooling cycle. All three originally-OWED CM- items CLOSED this session: `CM-SCRIPT-DOCS-SYNC` (commit `11820734`), `CM-COMMIT-DOCS-EXISTS` (commit `977630c3`), `CM-SUBAGENT-DELEGATION-AUDIT` (commit `2a0e11f4`). Each ships with standalone scanner + 7-or-8-fixture hermetic test +*

companion `docs/scripts/*.sh.md` user guide + wrapper integration + Bluff-Audit falsifiability rehearsal. The 5 Path-B equivalence-mapped items remain CLOSED-BY-EQUIVALENCE per §6.AD.3. Plus: T7 USB disk migration today (5G → 108G free on main; 7 dirs symlinked to T7 incl. ~/.gradle, ~/.cache, ~/.android, Xcode; `~/.zshrc` updated with `GRADLE_USER_HOME=/Volumes/T7/Gradle`, `XDG_CACHE_HOME`, `NPM_CONFIG_CACHE`).

Sweep tier-A closure (2026-05-17 evening, branch `sweep-findings-tier-A-2026-05-17`): 8 of the 10 comprehensive-sweep findings closed in a single coordinated commit (Findings #2 + #3 already closed by Bug 2 cascade + Bug 3 fix in prior cycles). All fixes falsifiability-rehearsed per §6.J / Seventh Law clause 1 (mutation applied → test fails with clear message → mutation reverted → test passes). Bluff-Audit stamps recorded in commit body.

- **Finding #1 (P0) — ToggleAnonymous persistence**: `feature/provider_config/.../ProviderConfigViewModel.kt` now persists via new `ProviderConfigRepository.setUseAnonymous(...)` → Room column `use_anonymous` (Migration 10→11 + schema 11.json). Switch state survives process restart.
- **Finding #4 (P1) — `LoginViewModel.serviceUnavailable` retype clear**: cleared in `validateUsername`/`validatePassword`/`validateCaptcha`/`onReloadCaptchaClick`/`onSubmitClick` reduces.
- **Finding #5 (P1) — `LoginViewModel.ServiceUnavailable` stale-captcha clear**: branch now sets `captcha = null, captchaInput = Initial` so the rendered challenge image doesn't lie when the sid expires.
- **Finding #6 (P1) — `ProviderLoginViewModel.serviceUnavailable` clear across selectProvider/backToProviders + retype**: symmetric fix to Findings #4/#5 on the multi-provider login surface.
- **Finding #7 (P1) — `OnboardingViewModel.onTestAndContinue` no more misleading "Invalid credentials"**: distinguishes `loginResult == null` (tracker has no auth path) from `loginResult.state != Authenticated` (real auth failure). Null path now treated as anonymous → switchTracker + advance.
- **Finding #8 (P1) — `OnboardingViewModel.loadProviders` excludes cloned synthetic trackers**: filters by syntheticId membership in `cloned_provider`. Clones remain configurable via Provider Config.
- **Finding #9 (P2) — `MainActivity` onboardingComplete re-read**: `PreferencesStorage.observeOnboardingComplete()` new Flow API (SharedPreferences-listener-backed). Two parallel `lifecycleScope.launch { repeatOnLifecycle { collect } }` blocks (one for theme, one for onboarding). Welcome screen re-appears if settings flip onboardingComplete back to false at runtime.
- **Finding #10 (P2) — `ToggleSync` first-tap race**: reads `toggleDao.get(providerId)?.enabled` synchronously instead of `state.syncEnabled`. First tap before observeAll() emit no longer silently flips the wrong direction.
- **Tests added** (5 new): `LoginViewModelTest` (3 cases: Findings #4 username, #4 password, #5 captcha), `ProviderConfigViewModelTest` (2 cases: Findings #1 + #10), Finding #7 + #8 cases added to existing `OnboardingViewModelTest`, Finding #6 case added to existing `ProviderLoginViewModelTest`.
- **Schema migration**: Room version bumped 10 → 11 (`MIGRATION_10_11` adds `use_anonymous INTEGER NOT NULL DEFAULT 0` to

provider_configs). core/database/schemas/lava.database.AppDatabase/11.json exported by KSP.
- **All builds + tests green**: :core:database / :core:credentials / :feature:provider_config /:feature

1.2.25-1045 distribute cycle (2026-05-17 afternoon):

- Stage-1 debug Firebase release ID 1lfjqc1nnhuio on digital.vasic.lava.client.dev
- Stage-2 release Firebase release ID 7p0h5j70eckqg on digital.vasic.lava.client (production)
- 15/15 Compose UI Challenge test cases PASS on Pixel_8/API35 host-direct AVD (14 classes including new C36)
- HelixConstitution submodule advanced to ca7c7d7 (§11.4.10.A Pre-store credential leak audit + upstream §11.4.37/38/39) — 5-mirror converged
- §6.L counter 57 → 58

Comprehensive UI/UX/core sweep findings (2026-05-17 evening, branch comprehensive-sweep-2026-05-17 merged at c3b8bf5c): 10 findings @ docs/sweeps/2026-05-17-comprehensive-uiux-core-sweep.md. Top P0s for 1.2.26:

1. ProviderConfigViewModel.ToggleAnonymous never persists — anonymous switch reverts on restart (feature/provider_config/.../ProviderConfigViewModel.kt:82-84)
2. SearchResultContent has no Error variant — Bug 2 root cause CONFIRMED ("all providers failed" looks like "0 results") (feature/search_result/.../SearchPageState.kt:30-54)
3. SearchInputViewModel.availableProviders is hardcoded 4-element list — SDK clones + new trackers invisible to search (feature/search_input/.../SearchInputViewModel.kt:48-53) P1 cluster (4 findings on login banner/captcha staleness + onboarding misleading cred message + clones-in-onboarding) + P2 cluster (MainActivity onboarding re-read + ToggleSync race). See sweep doc for full details + per-finding anti-bluff classification.

OPERATOR ACTION still REQUIRED:

- §6.H historical credential leak: rotate the RuTracker password (credentials remain valid until rotated; in-tree redaction does NOT purge git history per §6.T.3)
- Bug 2 live-device log capture: install 1.2.25, attempt anonymous-only search, adb logcat | grep -iE "error|exception|search" to confirm sweep finding #2 root cause

Bug 1 FULL REFACTOR LANDED — AuthResponseDto.ServiceUnavailable sealed variant propagates through 8-layer chain; Challenge C36 + 3 unit-layer falsifiability-rehearsed tests; OpenAPI spec + Go bindings regenerated; debug APK builds clean; feature/login + core/tracker/rutracker tests green

Bug 1 cycle (2026-05-17):

- Operator's §6.L 57th invocation forensic anchor "Cant login to RuTracker with valid credentials" closed. Partial fix in commit `17ceabcb` (stderr marker line) is now superseded by the full-fix variant: the SDK catch path returns `AuthResponseDto.ServiceUnavailable(reason)` instead of bluffing `WrongCredits(null)`.
- **8-layer propagation chain**: `AuthResponseDto` (+ variant) → `AuthMapper` → `AuthState.ServiceUnavailable(reason)` → `RuTrackerDtoMappers` (reverse) → `AuthServiceImpl` → `AuthResult.ServiceUnavailable` → `LoginUseCase` → `LoginResultMapper` → `ProviderLoginViewModel` (recordWarning telemetry per §6.AC + state field) → `ProviderLoginState.serviceUnavailable` → `ProviderLoginScreen` (renders "Service unavailable. Please try again later. (reason)" banner with `R.string.provider_login_service_unavailable` + `ServiceUnavailableTextTestTag`).
- Same wiring landed in legacy `LoginViewModel` + `LoginState.serviceUnavailable` for the single-tracker path.
- **OpenAPI spec**: `lava-api-go/api/openapi.yaml` gains `AuthResponseDtoServiceUnavailable` under `oneOf` + discriminator mapping. Go bindings regenerated via `scripts/generate.sh`; both `internal/gen/server/api.gen.go` + `internal/gen/client/api.gen.go` updated. Note: the Go-side rutracker scraper does NOT emit the new variant today — its login handler still returns Success / WrongCredits / CaptchaRequired only; the parity test holds. The variant exists in the spec so the Android Kotlin SDK wire-shape stays compatible with future Go-side adoption.
- **Tests added** (3 new + 1 rewritten):
  - `core/tracker/rutracker/src/test/kotlin/lava/tracker/rutracker/mapper/AuthMapperTest.kt` — added 2 tests for ServiceUnavailable forward mapping (with + without captcha).
  - `core/tracker/rutracker/src/test/kotlin/lava/tracker/rutracker/mapper/RuTrackerDtoMappersTest.kt` — added round-trip test asserting reverse-mapper preserves reason.
  - `core/tracker/rutracker/src/test/kotlin/lava/tracker/rutracker/impl/RuTrackerNetworkApiLoginUnknownRegressionTest.kt` — REWRITTEN: pre-fix asserted WrongCredits fallback (the §6.J bluff); now asserts ServiceUnavailable with reason carrying throwable class name. Same Crashlytics `a29412cf6566d0a71b06df416610be57` regression-immunity coverage, stronger discrimination.
  - `feature/login/src/test/kotlin/lava/login/LoginResultMapperTest.kt` — NEW file. 6 tests; load-bearing assertion: `service unavailable propagates as ServiceUnavailable with reason` + explicit anti-bluff `service unavailable does NOT silently collapse to WrongCredits`.
  - `feature/login/src/test/kotlin/lava/login/ProviderLoginViewModelTest.kt` — added VM test: `service unavailable shows banner does NOT mark creds Invalid`

`does NOT signal authorized` exercises the §6.J anti-bluff contract end-to-end at the VM layer (real ViewModel + real ProviderCredentialManager + real Room + real LavaTrackerSdk wired with FakeTrackerClient whose `loginProvider` returns `AuthState.ServiceUnavailable(reason)`).
    - `app/src/androidTest/kotlin/lava/app/challenges/Challenge36LoginServiceUnavailableShowsAccurateMessageTest.kt` — NEW Challenge Test under `// covers-feature: login`. Source-written + compiles green on darwin/arm64 (verified via `./gradlew :app:compileDebugAndroidTestKotlin`); EXECUTION against the §6.X-mounted gate-host is OWED.
- **Build verification**: `./gradlew :core:tracker:rutracker:test` PASS, `./gradlew :feature:login:test` PASS, `./gradlew :app:assembleDebug` PASS, `./gradlew :app:compileDebugAndroidTestKotlin` PASS. Spotless applied to all edited files.
- **§6.Y note**: this is a follow-up commit to 1.2.24-1044 — versionCode is NOT bumped here; the next distribute (1.2.25-1045) will pick up Bug 1 full fix + Bug 2 deferred investigation closure once that lands.
- **§6.X-debt**: C36 + the per-AVD matrix attestation row is OWED to a Linux x86_64 + KVM gate-host; the JVM-layer falsifiability rehearsals (4 tests) carry the gate until then.

Previous cycle context (preserved verbatim):

Last updated: 2026-05-16, **Phase 4-C-2 (pkg/detector adapter) + Phase 4-C-3 (pkg/ticket adapter) — BOTH WORKTREES LANDED + green; parent-cycle close + HelixQA SHA bump + coverage-ledger regen owed at meta-merge** (constitutional-plumbing-only; no user-visible feature change; no Firebase distribute since 1.2.22-1042 still serves the user-visible surface). Cycle spans commits `4def2da7` → `0c87b6ae` (33 commits since plan landing `832f739e`). Final state at HEAD `0c87b6ae`:

Major deliverables this cycle:

- The 12-clause constitution-compliance plan (`docs/plans/2026-05-15-constitution-compliance.md`) executed end-to-end across 10 phases. Plan + every phase's deliverable: see commit log between `832f739e..0c87b6ae`.
- HelixQA submodule incorporated (`submodules/helixqa` at upstream `b13ba7c0`) — see Phase 4 below.
- 40-gate verify-all sweep wrapper at full STRICT mode after Phase 7's coverage-ledger STRICT-flip.
- 17 own-org submodules now have helix-deps.yaml + install_upstreams.sh (16 vasic-digital + 1 HelixDevelopment HelixQA); 0 waivers in STRICT mode.
- §6.L counter advanced 36 → 52 across 17 back-to-back restatements (longest sequence in project history); 53rd in-flight per the dispatch that triggered this CONTINUATION.md refresh task.

Phase-by-phase status (constitution-compliance plan):

- ✅ **Phase 1** (§11.4.32 enforcement engine) — `4def2da7`. `scripts/verify-all-constitution-rules.sh` + meta-test + ci.sh wiring.

- ✅ **Phase 2** (§11.4.30 .gitignore audit gate) — `037389f5`. `scripts/check-gitignore-coverage.sh` + 16 new .gitignore files + sweep wiring + hermetic test.
- ✅ **Phase 3** (§11.4.31 helix-deps.yaml manifest gate) — `43345c3e` (gate) + `410af7ec` + `bcba3a19` (16/16 per-submodule manifests landed + Auth pin bump).
- ✅ **Phase 3-debt** CLOSED — all 16 vasic-digital submodules at pin advance with helix-deps.yaml present.
- ✅ **Phase 4** (§11.4.27 HelixQA + 100% test-type coverage) — `aa0db6bd`. HelixQA adopted as `submodules/helixqa` at upstream HEAD; `HELIX_DEV_OWNED` exemption pattern added to mirror-mandate scanners.
- ✅ **Phase 4 follow-up A** (Option 1 design) — `a61bd3d8`. Integration design at `docs/plans/2026-05-16-helixqa-integration-design.md` (Option 1 shell-wiring recommended; Options 2/3 deferred).
- ✅ **Phase 4 follow-up A executed** (shell-level wiring) — `1b66d192` + merge `d94ade0d`. 11 HelixQA Challenge scripts wrapped via `scripts/run-helixqa-challenges.sh` + `scripts/run-challenge-matrix.sh --include-helixqa` opt-in flag.
- ✅ **Phase 4 follow-up B** (4 open-question resolutions for Option 1) — `281780d7` + merge `84d871a5`. Runner-mode flag (`--runner=host|containerized`), toolchain-precondition gate (`HELIXQA_TOOLCHAIN_MAP`), evidence-dir env-var override, `HELIXQA_W_EXCLUSIONS` array consuming the §6.W audit doc. 11 hermetic fixtures in `tests/check-constitution/test_helixqa_wiring.sh`.
- 📐 **Phase 4 follow-up C** (DESIGN-ONLY) — `41b81359` + merge `be1ca3d8`. HelixQA Go-package linking design at `docs/plans/2026-05-16-helixqa-go-package-linking-design.md` (770-line Option 2 proposal: per-package adapters + 4-cycle rollout 4-C-1 `pkg/evidence` → 4-C-4 `pkg/navigator`+`pkg/validator`). **Operator-blocked on 10 open questions** (§G of the design); implementation cycle NOT started.
- ✅ **Phase 4-debt** CLOSED — `858ffb3e` (2026-05-16). HelixQA upstream PR `b13ba7c` landed `helix-deps.yaml` + `install_upstreams.sh` at the HelixQA repo root → Lava parent removed HelixQA from `HELIX_DEPS_WAIVERS` + `INSTALL_UPSTREAMS_WAIVERS`. 17/17 own-org submodules satisfy §11.4.31 + §11.4.35 + §11.4.36 in fully STRICT mode with **zero waivers**.
- ✅ **Phase 5** (§11.4.28 nested own-org submodule audit) — `bbca3a78` (gate) + `410af7ec` (STRICT flip after Challenges/.gitmodules removal via the github cascade merge).
- ✅ **Phase 5-debt** CLOSED — `410af7ec`. Scanner reports 0 violations in STRICT mode; Panoptic is no longer nested via Challenges.
- ✅ **Phase 6** (§11.4.29 lowercase snake_case naming) — `322f2081` (plan landing) + Phase 6a + 6b execution this cycle. Operator's 8 Q answers: defer Phase 6f upstream rename (Lava-side only); `helixqa` (single-token); `http3` (single-token); `ratelimiter` (single-token); Go `cmd/` hyphens exempt; ordering Mdns(low) → Containers(high) last; same defer for Tracker-SDK upstream; Phase 6a authorized to run in parallel with Phase 4-C-1. Execution: `Submodules/` → `submodules/` parent rename + 17 child renames (Auth→auth, Cache→cache, Challenges→challenges, Concurrency→concurrency, Config→config,

Containers→containers, Database→database, Discovery→discovery, HelixQA→helixqa, HTTP3→http3, Mdns→mdns, Middleware→middleware, Observability→observability, RateLimiter→ratelimiter, Recovery→recovery, Security→security, Tracker-SDK→tracker_sdk) + 139 referencing files updated + `.gitmodules` rewritten + Phase 6f upstream-rename execution plan at `docs/plans/2026-05-16-phase6f-upstream-rename-execution.md` (DEFERRED per Q1). Audit confirms 0 stale `Submodules/X` references in 16/17 names (Tracker-SDK has 1 ref in a historical bluff-hunt JSON narrative quote — exempt forensic anchor).
- ✅ **Phase 7** (§11.4.25 coverage ledger) — `21dee741` + merge `c35af27c` (generator + verifier + 58-row baseline + 6 hermetic fixtures + sweep wiring, advisory at first) → `76507ca0` + merge `20b3fd36` (waiver backfill: 0 covered / 20 partial / 38 gap → 48 covered / 10 partial / 0 gap) → `0c87b6ae` (STRICT-flip in sweep wrapper).
- ✅ **Phase 7-debt** CLOSED — `0c87b6ae`. Sweep wrapper now invokes `check-coverage-ledger.sh --strict` (was `--advisory`); gate hard-fails on stale/missing rows.
- ✅ **Phase 8** (§11.4.35 canonical-root + §11.4.36 install_upstreams) — `d95be689` (gate) + `410af7ec` (STRICT-flip after 10 install_upstreams scripts landed across owned submodules).
- ✅ **Phase 8-debt** CLOSED — `858ffb3e`. HelixQA's upstream `install_upstreams.sh` is the final hold-out; scanner reports 17/17 install_upstreams present in STRICT mode.
- ✅ **Phase 9 Path B** (§11.4.33 + §11.4.34 equivalence-mapping) — `055fbcbe`. §6.AD.3 amended: Lava's CONTINUATION + closure-logs + sixth-law-incidents satisfy the type-aware-closure + reopened-source-attribution semantics; no parallel Issues/Fixed tracker; equivalence is binding. Gates-index gets 2 new EQUIVALENCE-MAPPED rows (CM-CLOSURE-STATUS-VOCAB-COMPLIANCE, CM-REOPENED-SOURCE-ATTRIBUTION).

Verify-all sweep result at HEAD `0c87b6ae`: 40/40 PASS in fully STRICT mode (post-STRICT-flip; the prior 07:00:56Z attestation showed 39/40 only because the coverage-ledger sha drifted between commit and sweep — the re-run at HEAD is clean). Attestation directory: `.lava-ci-evidence/verify-all/<UTC-timestamp>.json`.

*CM- gates inventory at HEAD:** 24 HelixConstitution `CM-*` gates tracked at `docs/helix-constitution-gates.md`; ~16 wired + ~8 paper-only (mostly Issues/Fixed-tracker-dependent — equivalence-mapped per §6.AD.3). 14 Lava-side anti-bluff gates also active.

All 33 session commits §6.C-converged on GitHub + GitLab. Pre-push Checks 1-9 active throughout.

Prior: 2026-05-14, **1.2.23-1043 / 2.3.12-2312 closure-cycle** (constitutional-plumbing-only; no user-visible feature change). HelixConstitution submodule incorporated + §6.AD HelixConstitution-Inheritance Mandate landed; 8-track §6.AD-debt opened and systematically closed across 14 commits. §6.AC + §6.AB scanners in STRICT mode. Build-resource stats tracker (§11.4.24) landed. All Lava-side debts in scope CLOSED at commit `4a7d0402`. See git log `66de343b..4a7d0402` for the full closure-cycle.

> **Prior:** 2026-05-14, **1.2.22-1042 / 2.3.11-2311 DISTRIBUTED to Firebase** (debug stage 1 + release stage 2, operator pre-authorized combined). About dialog author re-order; Crashlytics 6-issue sweep (3 fixed + 3 closed-historical); §6.AC Comprehensive Non-Fatal Telemetry Mandate added (28th §6.L); §6.AA-debt PARTIAL CLOSE; per-channel last-version-{debug,release} pointers; both APIs running 2.3.11.
>
> **Prior:** 2026-05-14, 1.2.21-1041 DISTRIBUTED (onboarding back-press fix + WelcomeStep colored Image fix; §6.AB Anti-Bluff Test-Suite Reinforcement, 27th §6.L).
>
> **Prior:** 2026-05-14, 1.2.20-1040 DISTRIBUTED (Galaxy S23 Ultra cold-launch crash fix: `ic_lava_logo` layer-list → composited PNG; §6.Z Anti-Bluff Distribute Guard, 26th §6.L).
>
> **Prior:** 2026-05-14, 1.2.19-1039 DISTRIBUTED (§6.Y Post-Distribution Version Bump Mandate, 25th §6.L) + 1.2.18-1038 DISTRIBUTED (24th §6.L).
>
> **Prior:** 2026-05-13, SP-4 Phase G + Phase F.1 + F.2 + Phase D (multi-provider parallel search SDK) + Phase C (Trackers screen removal + :feature:provider_config landing). See git log `0c87b6ae..` and the CHANGELOG for the full delivery chain.
>
> **§6.S binding:** this file is constitutionally load-bearing per root `CLAUDE.md` §6.S. Every commit that changes phase status, lands a new spec/plan, bumps a submodule pin, ships a release artifact, discovers or resolves a known issue, or implements an operator scope directive MUST update this file in the SAME COMMIT. The §0 "Last updated" line MUST track HEAD. Stale CONTINUATION.md is itself a §6.J spirit issue under §6.L's repeated mandate. `scripts/check-constitution.sh` enforces presence + structure (§0, §7, §6.S clause + inheritance).

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0. Quick orientation (read this first)

Surface	Current state	Pin
Lava parent on master	2 mirrors (GitHub + GitLab) converged at HEAD	§6.L 68th cycle (see `git log` for current SHA)
API (lava-api-go)	2.3.22 (code 2322) — `internal/version/version.go`	container `lava-api-go-thinker`
Android Firebase	1.2.33 (1053) distributed to testers (2026-05-18, last user-visible release; `last-version-{debug,release}` both = 1053)	`lava-vasic-digital` Firebase project
17 own-org submodules	all pushed (16 vasic-digital + 1 HelixDevelopment HelixQA)	see §3
constitution submodule	at upstream HEAD `883ccc1` (§11.4.79–§11.4.106 adopted via §6.AF)	HelixDevelopment/HelixConstitution

Surface	Current state	Pin
Workable-items tracker	canonical `workable-items` binary + `docs/workable_items.db` (tracked, 8 items LVA-1..8; LVA-3 migrated, LVA-tickets retired)	§11.4.93/95/106 + §11.4.74
codegraph	incorporated 2026-05-20 (§11.4.78); local SQLite index at `.codegraph/` (1,182 files / 18,567 nodes)	`@colbymchenry/codegraph` MCP
Verify-all sweep	40/40 PASS, fully STRICT mode (last attested prior cycle; 68th-cycle re-run in progress)	`.lava-ci-evidence/verify-all/`
Coverage ledger	48 covered / 10 partial / 0 gap (58 rows)	`docs/coverage-ledger.yaml`
CM-* gates wired	~16 of 24 wired (8 paper-only or equivalence-mapped)	`docs/helix-constitution-gates.md`

This cycle delivered the entire 12-clause constitution-compliance plan plus HelixQA submodule adoption plus the Phase 7 STRICT-flip. **No user-visible feature change**; constitutional-plumbing-only.

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1. What's DONE (this cycle, since 2026-05-15)

Constitution-compliance plan (`docs/plans/2026-05-15-constitution-compliance.md`)

Phase	Subject	Status	Anchor commits
0	Pin advance + plan land	✅ DONE	`ed16debd` (pin) + `832f739e` (plan)
1	§11.4.32 enforcement engine	✅ DONE	`4def2da7`
2	§11.4.30 .gitignore audit gate	✅ DONE	`037389f5`
3	§11.4.31 helix-deps.yaml manifests	✅ DONE	`43345c3e` + `410af7ec` + `bcba3a19`
3-debt	16/16 per-submodule manifests	✅ CLOSED	`410af7ec` + per-submodule pin advances
4	§11.4.27 HelixQA + 100% test-type coverage	✅ DONE	`aa0db6bd`
4 follow-up A	Option 1 design + shell wiring	✅ DONE	`a61bd3d8` + `1b66d192` + merge `d94ade0d`
4 follow-up B	4 open-question resolutions	✅ DONE	`281780d7` + merge `84d871a5`

Phase	Subject	Status	Anchor commits
4 follow-up C	HelixQA Go-package linking design	📐 DESIGN-ONLY	`41b81359` + merge `be1ca3d8`
4-C-1	Lava-side `pkg/evidence` adapter (WRAP)	✅ DONE 2026-05-16	HelixQA `a1e2020d` + Lava `573b4a8a`
4-C-2	Lava-side `pkg/detector` adapter (WRAP)	✅ DONE 2026-05-16	HelixQA `a1e2020d` unchanged + Lava `<this-commit>`
4-debt	HelixQA upstream install_upstreams.sh + helix-deps.yaml	✅ CLOSED 2026-05-16	`858ffb3e`
5	§11.4.28 nested-own-org submodule audit	✅ DONE	`bbca3a78`
5-debt	STRICT flip after Challenges/.gitmodules removal	✅ CLOSED	`410af7ec`
6	§11.4.29 lowercase snake_case naming	📐 PLAN-ONLY	`322f2081` + merge `c8d42434`
7	§11.4.25 coverage ledger	✅ DONE	`21dee741` + merge `c35af27c`
7-debt	waiver backfill + STRICT flip	✅ CLOSED	`76507ca0` + merge `20b3fd36` + `0c87b6ae`
8	§11.4.35 + §11.4.36 canonical-root + install_upstreams	✅ DONE	`d95be689` + `410af7ec`
8-debt	10 install_upstreams scripts across owned submodules	✅ CLOSED	`410af7ec`
9 Path B	§11.4.33 + §11.4.34 equivalence-mapping	✅ DONE	`055fbcbe`

HelixQA submodule (NEW this cycle)

- Adopted at `submodules/helixqa` from `git@github.com:HelixDevelopment/HelixQA.git`.
- Initial pin: `403603db` (2026-05-15 in Phase 4).
- Upstream PR `b13ba7c` added `helix-deps.yaml` + `install_upstreams.sh` (2026-05-16 in Phase 4-debt closure).
- Current pin: `b13ba7c0`.
- 11 HelixQA Challenge scripts wrapped via `scripts/run-helixqa-challenges.sh` (Option 1 shell-level wiring).
- 11 hermetic fixtures at `tests/check-constitution/test_helixqa_wiring.sh` validate the wrapper.
- §6.W audit doc at `docs/helixqa-script-audit.md` (per-script git-push / curl / outside-worktree analysis; 0/11 violators on default config).
- `HELIX_DEV_OWNED` exemption pattern in `scripts/check-canonical-root-and-upstreams.sh` + `scripts/check-helix-deps-manifest.sh`

(HelixDevelopment org submodules treated like vasic-digital for mirror-presence checks, distinct from arbitrary third-party submodules).

§6.L counter advance

The §6.L Anti-Bluff Functional Reality Mandate counter advanced from 36 to 52 across the cycle (53rd is in-flight per the dispatch that triggered this CONTINUATION.md refresh task). 17-cycle back-to-back restatement is the longest sequence in project history; per §6.L the repetition itself is the constitutional record. Anchor commits: `8c47cd17`, `d159d0fc` (37-41 batched), `66803d4d` (43+44 batched), `aa0db6bd` (45), `a61bd3d8` (46), `ed7a658d` (47), `dcec9eb8` (48), `0f1b19f1` (49), `2882304b` (50+51), `0c87b6ae` (52).

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2. What's BLOCKED ON OPERATOR ACTION

These items need the operator's environment / hardware / decisions that an agent cannot make alone.

2.1 Phase 4 follow-up C (HelixQA Go-package linking) — STATUS UPDATE 2026-05-16

Phase 4-C-1 (pkg/evidence adapter): COMPLETED 2026-05-16. All 10 open questions answered by operator + implementation landed in this cycle's commit. Operator decisions: Q1 Go 1.26 bump, Q2 WRAP, Q3 Path A tag-pin (transitional Path B replace+sibling-mount until HelixQA stabilizes), Q4 preserve HelixQA terminology (Collector / Detector / Generator), Q5 upstream-contribute CaptureGeneric first (HelixDevelopment/HelixQA PR #1, branch `feat/evidence-capture-generic`, commit `a1e2020dd759d025b67ef8e024061b103940470d`), Q6 SKIP 4-C-4 navigator entirely, Q7 NO recover() wrapping, Q8 accept 2x CI build-time delta, Q9 always-track-upstream for HelixQA (§6.AD waiver documented in CLAUDE.md), Q10 coverage-ledger bumped in same commit.

Deliverables this cycle:

- HelixQA `pkg/evidence.CaptureGeneric` public method (commit `a1e2020d`, PR HelixDevelopment/HelixQA#1)
- `lava-api-go/internal/qa/evidence/{collector,collector_test}.go` (adapter + 9 unit tests, 87.9% coverage)
- `lava-api-go/tests/qa/evidence_test.go` (real-stack integration test, `//go:build helixqa_realstack`)
- `lava-api-go/go.mod` bumped to Go 1.26 + adds `digital.vasic.helixqa` require + replace
- `Submodules/HelixQA/` pin bumped to `a1e2020d`
- `docs/coverage-ledger.yaml` regenerated (58 rows; lava-api-go row: 89 unit tests + 1 integration)
- `CLAUDE.md` §6.AD-debt: HelixQA always-track-upstream waiver documented
- This `CONTINUATION.md` updated

Phase 4-C-2 (detector adapter), 4-C-3 (ticket adapter), 4-C-4 (validator + SKIP navigator per Q6) remain owed — each is 1-session scope per design doc §E.

Phase 4-C-2 (pkg/detector adapter): COMPLETED 2026-05-16. Operator decisions reused from 4-C-1 (Q1–Q10 unchanged). Q4 preserves `Detector` name. Q5: no HelixQA-side promotion needed — `pkg/detector`'s public API surface already exposes everything the adapter requires (`Detector`, `Option`, `New`, `WithDevice`/`WithPackageName`/`WithBrowserURL`/`WithProcessName`/`WithProcessPID`/`WithEvidenceDir`/`WithCommandRunner`, `Check`, `CheckApp`, `Platform`, `DetectionResult`, `CommandRunner` interface).

Deliverables this cycle:

- `lava-api-go/internal/qa/detector/detector.go` (255 LOC) — WRAP-strategy adapter exposing Lava-shaped `Report` struct (Crashed/Alive/StackTrace/EvidencePath); `CheckGoProcess(processName)` for name-based detection (real `pgrep -f`); `CheckGoProcessByPID(pid)` for PID-based detection (real `kill -0`); `ErrEmptyProcessName` + `ErrInvalidPID` Lava-side guards that block HelixQA's silent fallback to `processName="java"` for empty/zero inputs (forensic value: mutation rehearsal proved Alive=true would silently be returned because Java exists on every dev box)
- `lava-api-go/internal/qa/detector/detector_test.go` (11 tests, 82.9% statement coverage, race-clean) — uses HelixQA's `CommandRunner` interface as the boundary-fake (not a mock of the SUT; the HelixQA Detector itself runs unaltered, satisfying §6.J.4 forbidden-mock pattern)
- `lava-api-go/tests/qa/detector_test.go` (3 real-stack tests, `//go:build helixqa_realstack`) — spawns sacrificial `sleep` child processes via `sh -c 'exec -a <sentinel> sleep 30'` (BSD/macOS-compatible argv[0] injection); asserts the REAL HelixQA Detector against the real OS process table
- `docs/coverage-ledger.yaml` regenerated (lava-api-go unit_test_count 89 → 93)
- This `CONTINUATION.md` updated; coverage-ledger regen confirmed (58 rows preserved)

§6.J anti-bluff posture: 4 falsifiability rehearsals captured in commit body — (1) `Crashed: dr.HasCrash → !dr.HasCrash` triggered 5 unit-test assertions; (2) empty-name guard removal caught at compile time (`"strings" imported and not used`); (3) PID guard removal triggered 2 sub-test assertions + exposed HelixQA's silent `java` fallback; (4) `Alive: dr.ProcessAlive → !dr.ProcessAlive` triggered ALL 3 real-stack assertions on live/dead/ghost process paths. All reverted, all green after revert.

Honest scope statement: real-stack tests were initially executed via `go test -tags=helixqa_realstack ./tests/qa/detector_test.go` (file-target form) while Phase 4-C-3's `internal/qa/ticket/generator.go` was untracked + uncompiling. After 4-C-3 landed at `86402bfa` (which also bumped HelixQA pin `a1e2020d` → `c57c275` to gate `enhanced_generator.go` behind the `helixqa_enhanced_tickets` tag), the full package-target form `go test -tags=helixqa_realstack ./tests/qa/...` PASSes too (7/7: 2 evidence + 3 detector + 2 ticket). Re-verified at HEAD post-`86402bfa`.

Phase 4-C-3 (pkg/ticket adapter): COMPLETED 2026-05-16. Operator decisions reused from 4-C-1 (Q1–Q10 unchanged). Q4 preserves `Generator` name.

Deliverables this cycle:

- HelixQA-side prereq: `submodules/helixqa/pkg/ticket/enhanced_generator.go` gated behind `//go:build helixqa_enhanced_tickets` so plain consumers of `pkg/ticket` (the Lava adapter) do NOT pull LLMOrchestrator transitively. HelixQA SHA bump owed at parent-cycle close.
- `lava-api-go/internal/qa/ticket/generator.go` (≈340 LOC) — WRAP-strategy adapter with `NewGenerator`, `GenerateClosureLog`, `OutputDir`, `ClosureLogInput` shape mirroring §6.O closure-log conventions
- `lava-api-go/internal/qa/ticket/generator_test.go` (13 tests + 8 sub-tests, 93.2% statement coverage with `-race`)
- `lava-api-go/tests/qa/ticket_test.go` (2 real-stack tests, `//go:build helixqa_realstack`)
- `CLAUDE.md` §6.O extended with clause 7 — adapter authorized as programmatic closure-log path
- This `CONTINUATION.md` updated; coverage-ledger row addition owed at parent-cycle rolled regen

§6.J anti-bluff posture: 2 falsifiability rehearsals captured in commit body — (1) H1 heading mutation surfaced by schema test, (2) empty-CrashlyticsID validation removal surfaced by RejectsEmpty test; both reverted, both green after revert.

Phase 4-C-4 (validator adapter; navigator SKIPPED per Q6) remains owed.

2.2 Phase 6 snake_case migration — RESOLVED + EXECUTED

All 8 operator questions answered (2026-05-16). Phase 6a + 6b executed in this cycle. See `docs/plans/2026-05-16-phase6f-upstream-rename-execution.md` for the deferred upstream-rename plan (Q1: defer; document execution steps for operator).

2.3 Release-tagging chain (versions inherit from prior cycle)

Last Firebase distribute: 1.2.22-1042 / 2.3.11-2311 (2026-05-14). This cycle made NO user-visible changes, so no new distribute is owed. Tag-script gate per §6.I (multi-emulator container matrix + per-AVD attestation) still blocked on Linux x86_64 + KVM gate-host per the standing §6.X-debt (`.lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json`).

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3. Submodule pin index

17 own-org submodules + 1 universal-rules submodule (constitution). Bumping a pin is a deliberate operator action; never auto-update.

Submodule	Pin	Mirrors	Notes
`auth`	`24ca50db`	GitHub + GitLab	helix-deps.yaml + §11.4.78 CodeGraph cascade

Submodule	Pin	Mirrors	Notes
`cache`	`30bb8581`	GitHub + GitLab	helix-deps.yaml present
`challenges`	`09c55f48`	GitHub + GitLab	helix-deps.yaml + flat layout (Panoptic dep declared)
`concurrency`	`7c74625b`	GitHub + GitLab	helix-deps.yaml present
`config`	`9491f8b4`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`containers`	`6aff7ea8`	GitHub + GitLab	helix-deps.yaml + per-OS emulator acceleration (§6.X-debt darwin/arm64 RESOLVED)
`database`	`4ead6233`	GitHub + GitLab	helix-deps.yaml present
`discovery`	`2bddf64c`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`helixqa`	`8b12e922`	GitHub	HelixDevelopment org; always-track-upstream per §6.AD Q9 waiver
`http3`	`1d0df7b7`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`mdns`	`ba1d2385`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`middleware`	`6ee9c0ec`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`observability`	`2b8c1633`	GitHub + GitLab	helix-deps.yaml present
`ratelimiter`	`9da442cb`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`recovery`	`58f9b4f9`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`security`	`a388cc44`	GitHub + GitLab	helix-deps.yaml present
`tracker_sdk`	`7afc37aa`	GitHub + GitLab	helix-deps.yaml + install_upstreams.sh
`constitution`	`208e2c8`	universal (HelixConstitution upstream)	adds §11.4.78 CodeGraph mandate

Internal-to-submodule nested submodules: `submodules/challenges` had a nested `Panoptic` submodule that was removed via the github cascade merge in Phase 5-debt closure (CONST-051(C) flat-layout enforcement); Challenges now declares Panoptic as a `layout: flat` dependency in its `helix-deps.yaml`.

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4. Known issues + bugs (carried forward — historical)

These are real defects discovered before this cycle. Tracked here for forensic continuity; none are blocking this cycle's constitutional work.

4.5 Active known issues

- **§6.X-debt (Linux x86_64 + KVM containerized gate path)**: STANDING for the Linux host path only. The **darwin/arm64 sub-debt is RESOLVED** (2026-05-20, commit `23c508e9`): per-OS emulator acceleration (`AccelProfileForOS` / `ResolveRunner` / `GateEligibleForOS` + `emulator-matrix --runner=auto` in Containers `c1871138`+`6aff7ea8`) makes the macOS gate runner host-direct+HVF (a Linux container cannot reach the host-only HVF API). PROVEN: C00 cold-start canary + full 37-class Challenge suite on Pixel_8/API35 = 43 pass / 3 credential-skip / 0 fail. The Linux x86_64 containerized-KVM path remains owed. Forensic anchor: `.lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json`.
- **§6.H Firebase CI token echo-leak** (2026-05-20, §6.L 67th): **RESOLVED 2026-05-31** — operator rotated the token (`firebase logout` → `firebase login:ci`) during the §6.L 68th cycle; the transcript-leaked token (never committed to git) is now dead. §6.H clause 6 satisfied. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-05-20-firebase-token-echo-leak.json`.
- **LVA-8 — HelixQA crash-detector + consumer fixture** — **RESOLVED 2026-05-31** (§6.L 68th): `internal/qa/validator` has 6 failing tests because HelixQA's `isPIDAlive` (`submodules/helixqa/pkg/detector/desktop.go`) shells `exec kill -0 <pid>` and `/bin/kill -0 <absent-pid>` returns EXIT 0 on macOS (bash builtin returns 1), so a dead PID reads as alive → Validator reports StepPassed on a crashed step (the canonical §6.J bluff). ROOT CAUSE CONFIRMED (captured evidence). Fix belongs UPSTREAM in HelixQA (use `syscall.Kill` not `/bin/kill`) per CONST-051 + CONST-049. Incident: `.lava-ci-evidence/sixth-law-incidents/2026-05-31-helixqa-validator-killbinary-macos-bluff.json`. The Lava adapter is a faithful pass-through (0-byte `internal/` diff) — the defect is entirely HelixQA-side. **Operator decision owed:** authorize the HelixQA upstream fix cycle (fix → push to HelixQA → bump pin).
- **LVA-3 — LVA-vs-canonical-workable-items reconciliation** (§6.L 68th): the constitution ships a canonical `workable-items` Go binary at `constitution/scripts/workable-items/` keyed `docs/workable_items.db`; Lava built a parallel LVA-keyed system at `tools/lava-tickets/` + `docs/tickets/tickets.db`. Both satisfy §11.4.93/95/106; whether LVA supersedes or complements the canonical binary (§11.4.74 catalogue-first) is an **operator decision**. Until ratified both the LVA system AND the §6.AD.3 Path-B `.lava-ci-evidence/` ledgers stay in force.
- **macOS emulator stall** (2026-05-15 incident): Pixel_7_Pro on macOS
  - emulator 36.1.9 stalls indefinitely. Three candidate root-causes recorded as `PENDING_FORENSICS:` (T7 external drive contention, emulator-36.1.9 known issues, AVD config theory eliminated by fresh-AVD re-test). Forensic anchor: `.lava-ci-evidence/sixth-law-incidents/2026-05-15-macos-emulator-stall-on-android33.json`. Orthogonal to §6.X-debt.

- **github SSH-fail flake pattern** (resolved this cycle): the §6.L 37th
    - 39th invocation forensics document multi-push retry pattern; the resolution is the standing operator practice of retrying `git push github` on connection-reset. No code change owed.

4.5 Resolved this cycle

- **Flaky `CredentialsViewModelTest > select provider updates selectedProvider`** (§6.L 68th, 2026-05-31): the fixed-`awaitState()`-count assumption was replaced with a bounded await-until-`selectedProvider=="rutracker"` loop, removing the dependence on the non-deterministic interleaving of `load()`'s Room-`Flow.first()` resume (delivered off the StandardTestDispatcher) vs. the `SelectProvider` reduce. Falsifiability-rehearsed (broke the reduce → `AssertionFailedError: expected:<rutracker> but was:<null>`, localized to that one test → reverted → 6/6 green). Incident JSON `.lava-ci-evidence/sixth-law-incidents/2026-05-20-flaky-credentialsviewmodeltest.json` remains as the forensic record.
- **`.codegraph/*.pid` gitignore gap** (§6.L 68th): `daemon.pid` was untracked-but-not-ignored (§11.4.30); `.codegraph/*.pid` added to `.gitignore`.
- **Stale coverage ledger (§11.4.25)** (§6.L 68th): the verify-all sweep was 46/47 — `docs/coverage-ledger.yaml` had drifted (prior cycles added `feature/login` LoginViewModelTest, `feature/onboarding` AuthTypeDisplayTest + challenge C26, `feature/provider_config` ProviderConfigViewModelTest without regenerating the ledger). Regenerated via `scripts/generate-coverage-ledger.sh`; `check-coverage-ledger.sh --strict` now EXIT=0. Sweep → **47/47 PASS**.

LVA ticket system NEW (§6.L 68th, §11.4.93/95/106)

The operator's "define ticket key LVA + SQLite workable-items DB + Issues/Fixed/Issues_Summary/Fixed_Summary + PDF/HTML/DOCX exports" directive is the same requirement as the new HelixConstitution **§11.4.93/95/106** (workable-items SQLite DB tracked-in-git + mechanical md⍰DB byte-identical sync). Built as a pure-Go module at `tools/lava-tickets/` (`digital.vasic.lava.tickets`, Go 1.26, `modernc.org/sqlite` — no CGO, no sudo). Key prefix **`LVA`** (LVA-1, LVA-2, …; the Lava instantiation of §11.4.54 ATM-NNN). The DB `docs/tickets/tickets.db` **IS tracked** (§11.4.95 — never gitignored); only `*.db-wal/-shm/-journal` sidecars + `bin/` + scratch are ignored. Subcommands: `init/add/update/close/reopen/gen/verify/import/export/list/version`. `go test` 7/7 PASS (incl. §11.4.106 round-trip + falsifiability, §11.4.33 type-aware closure, §11.4.34 reopen-attribution); `verify` byte-identical PASS (exits 1 on drift — independently confirmed). Exports: HTML (pure-Go) ✓, DOCX (podman `pandoc/core`) ✓, **PDF honestly BLOCKED** (container lacks a LaTeX engine — tool exits 3, writes NO fake file; remediation: `pandoc/latex` image or host `weasyprint`). 7 real LVA tickets seeded from actual project state (no invented SHAs). Design + verbatim build evidence: `docs/tickets/DESIGN.md` + `docs/tickets/BUILD-EVIDENCE.md`. **OPEN (operator decision, LVA-3):** whether the LVA system supersedes or complements the §6.AD.3 Path-B `.lava-ci-evidence/` ledgers — until ratified, the Path-B mapping remains the binding compliance surface and the LVA DB is seeded from it (reconciled, not replacing).

Constitution pin BUMPED `208e2c8` → `883ccc1` (§6.L 68th, §6.AF)

Operator directed bump-now+adopt. 53 upstream commits (2026-05-20 → 2026-05-31) add **§11.4.79–§11.4.106** (28 universal clauses). 68th-cycle review at `.lava-ci-evidence/constitution-review/2026-05-31-68th-cycle-review.md`: **NO constitution-mandated submodule is missing** (Challenges, HelixQA, containers, codegraph all present). New §6.AF clause in CLAUDE.md enumerates per-clause Lava adoption status + §6.AF-debt for the OWED items:

- §11.4.93/95/106 (workable-items DB) — **SATISFIED** by the LVA system. The constitution also ships a canonical `workable-items` Go binary at `constitution/scripts/workable-items/` keyed `docs/workable_items.db`; LVA-vs-canonical reconciliation is operator-gated (LVA-3).
- §11.4.79/.80 (own-org submodules IN codegraph index) — **OWED** (LVA-6; Lava currently excludes `submodules/`).
- §11.4.85 (stress+chaos) — **IN PROGRESS** (LVA-7; phase-1 lava-api-go scaffold + evidence under `docs/chaos-stress/`).
- operating-mode clauses — EQUIVALENCE-MAPPED to existing Lava practice. §11.4.100 (video-color) DEMOTED to ATMOSphere-only — not binding on Lava. The constitution submodule pin bump is a parent-repo change; per CONST-049 the constitution stays pinned + advanced deliberately (NOT auto-tracking).

Resolved in prior (constitution-compliance) cycle

- **Ledger-staleness drift class** — Phase 7-debt closure (`0c87b6ae`) flipped the coverage-ledger gate from `--advisory` to `--strict` in the sweep wrapper. Subsequent stale-ledger commits will hard-fail at sweep time + pre-push.
- **§11.4.27 HelixQA non-incorporation** — Phase 4 closure (`aa0db6bd`)
  - Phase 4-debt closure (`858ffb3e`) bring HelixQA in as `submodules/helixqa` with full mirror compliance.
- **§11.4.31 / .35 / .36 zero-waiver state** — Phase 4-debt closure + Phase 8-debt closure achieve 17/17 own-org submodules satisfying all three mandates with **zero waivers** in STRICT mode.

4.5 Historical — pre-this-cycle, carried forward

(See full historical detail in the git log between the prior CONTINUATION.md "Last updated" header and `4a7d0402`. Summary: C02 Cloudflare-mitigation stops short of profile-parsing; C17-C22 require emulator matrix; UDP buffer warning documented; mirror model reduced to 2-mirror per §6.W; docs/todos/Lava_TODOs_001.md committed as historical; etc.)

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5. Operator-flagged follow-up items (small, queued)

- **Phase 4-C implementation** — blocked on §2.1 open questions.
- **Phase 6a implementation** — blocked on §2.2 open questions.
- **HelixQA pin upgrade cadence** — operator decides when to re-baseline to track HelixQA `main` vs. holding at `b13ba7c0`.

- **Re-audit HelixQA scripts (`docs/helixqa-script-audit.md`)** on every pin bump per the §6.W audit doc's re-audit-trigger clause.
- **Coverage-ledger row additions** when new feature modules land — the generator is deterministic; re-run via `scripts/generate-coverage-ledger.sh` in the same commit as the new module to keep STRICT mode green.

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6. Constitutional debt + memory anchors

- **§6.K-debt** (Containers extension): RESOLVED 2026-05-07.
- **§6.N-debt** (pre-push hook enforcement): RESOLVED 2026-05-05.
- **§6.AD-debt** (HelixConstitution-Inheritance per-scope + CM-* wiring): FULLY DRAINED 2026-05-18. Of the original `CM-*` set: 5 mapped Path-B-equivalent (CLOSED-BY-EQUIVALENCE per §6.AD.3), `CM-UNIVERSAL-VS-PROJECT-CLASSIFICATION` CLOSED by pre-push Check 8, `CM-SCRIPT-DOCS-SYNC` CLOSED 2026-05-17 (`11820734`), `CM-COMMIT-DOCS-EXISTS` CLOSED 2026-05-18 (`977630c3`), `CM-SUBAGENT-DELEGATION-AUDIT` CLOSED 2026-05-18 (`2a0e11f4`). No items remain OWED.
- **§6.X-debt** (Linux x86_64 + KVM gate-host for container-bound emulator matrix): STANDING. See §4.5 above.
- **§6.L** (Anti-Bluff Functional Reality Mandate): 52 invocations across multiple working days; 17-cycle back-to-back the longest sequence in project history this cycle. Per §6.L the repetition IS the constitutional record.
- **§6.R** (No-Hardcoding Mandate): UUID + IPv4 + host:port scanners active; algorithm-parameter literal grep staged (code-review gate per §6.R clause body).
- **§6.S** (Continuation Document Maintenance): THIS file. Per §6.S the §0 "Last updated" line MUST track HEAD.
- **§6.T** (Universal Quality Constraints): four sub-points active (Reproduction-Before-Fix, Resource Limits, No-Force-Push, Bugfix Documentation).
- **§6.AC** (Comprehensive Non-Fatal Telemetry): scanner in STRICT mode; ci.sh hard-fail wired.
- **§6.AB** (Anti-Bluff Test-Suite Reinforcement): scanner in STRICT mode.
- **§6.AE** (Comprehensive Challenge Coverage + Container/QEMU Matrix): per-feature scanner in STRICT mode; container matrix runner BLOCKED on §6.X-debt.

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7. RESUME PROMPT

Paste the following into a new CLI agent session to continue this work. The agent needs no scrollback — everything it needs is in this file plus the spec/plan/CLAUDE.md set referenced from it.

```
Continue Lava project work. Read these in order before doing
anything:

  1. /Users/milosvasic/Projects/Lava/docs/CONTINUATION.md
  2. /Users/milosvasic/Projects/Lava/CLAUDE.md
  3. /Users/milosvasic/Projects/Lava/constitution/Constitution.md
```

  4. /Users/milosvasic/Projects/Lava/docs/plans/2026-05-15-constitution-compliance.md
  5. /Users/milosvasic/Projects/Lava/docs/helix-constitution-gates.md
  6. /Users/milosvasic/Projects/Lava/docs/coverage-ledger.yaml (skim — generated)

Then check the git state vs the CONTINUATION.md "Last updated" line.
If new commits exist on master beyond what CONTINUATION.md describes,
trust the commits and update CONTINUATION.md before proceeding (per §6.S).

Active state per CONTINUATION.md §1 (2026-05-16):
  - All 10 phases of the constitution-compliance plan DONE (Phases 1-9 closed).
  - HelixQA submodule incorporated; Phase 4-debt CLOSED 2026-05-16.
  - Verify-all sweep: 40/40 PASS in fully STRICT mode.
  - 17/17 own-org submodules with helix-deps.yaml + install_upstreams.sh; zero waivers.
  - Coverage ledger: 48 covered / 10 partial / 0 gap (58 rows).
  - §6.L counter at 52; 53rd in-flight at the moment of this CONTINUATION.md refresh.
  - 33 session commits §6.C-converged on GitHub + GitLab.

Your default next action (priority order):
  1. **Phase 4-C** (HelixQA Go-package linking): blocked on 10 operator open
     questions at `docs/plans/2026-05-16-helixqa-go-package-linking-design.md`
     §G. Surface the questions to the operator; do NOT proceed to 4-C-1.
  2. **Phase 6a** (snake_case migration): blocked on 8 operator open questions
     at `docs/plans/2026-05-16-snake_case-migration.md` §11. Surface the
     questions to the operator; do NOT proceed to Phase 6a implementation.
  3. **Crashlytics monitoring**: the last Firebase distribute was 1.2.22-1042
     (2026-05-14); check Crashlytics for any new issues per §6.O closure
     mandate.
  4. **HelixQA pin freshness**: re-baseline `submodules/helixqa` from
     upstream if operator approves; re-run the §6.W audit.
  5. **Tag-script gate**: still blocked on §6.X-debt (Linux x86_64 + KVM
     gate-host) for §6.I matrix attestation. No release tag this cycle.

Do NOT re-run completed phases — they are committed + pushed +

```
sweep-verified.
The git log is the authoritative record.

Verify-all sweep evidence: `.lava-ci-evidence/verify-all/<UTC-timestamp>.json`
Latest gates index: `docs/helix-constitution-gates.md`
Coverage ledger: `docs/coverage-ledger.yaml`

Constitutional bindings still in force (do not relax):
  §6.J / §6.L (Anti-Bluff Functional Reality Mandate)
  §6.AB / §6.AC / §6.AE (anti-bluff scanners — STRICT)
  §6.AD (HelixConstitution Inheritance) + §6.AD.3 (equivalence-mapping)
  §6.R (No-Hardcoding Mandate)
  §6.S (Continuation Document Maintenance — THIS file)
  §6.W (GitHub + GitLab Only Remote Mandate; HELIX_DEV_OWNED exemption for HelixDevelopment org)
  §6.X (Container-Submodule Emulator Wiring; PARTIAL — gate-host owed)
  §11.4.25–§11.4.36 (12 new HelixConstitution clauses)

The operator's standing §6.L wall is preserved verbatim in CLAUDE.md.
Read it.
```

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8. House-keeping the agent should keep doing

These are habits established across multiple cycles; future agents should preserve them.

1. **Commit messages carry Bluff-Audit stamps for every test class added or modified** (Seventh Law clause 1; pre-push Check 2 rejects commits without them).
2. **Every commit must have `Co-Authored-By: Claude Opus 4.7 (1M context) <noreply@anthropic.com>`** as the trailer.
3. **Push to both Lava parent mirrors (GitHub + GitLab)** after every commit chain that closes a logical unit. After every push, confirm convergence with `for r in github gitlab; do echo "$r: $(git ls-remote $r master | awk '{print $1}' | head -1)"; done`.
4. **Submodule pushes are explicit per submodule** to whatever remotes that submodule has (varies — see §3). Never use `git submodule foreach git push` blindly.
5. **Update this CONTINUATION.md** in the same commit as any completion-state change (phase done, new spec/plan written, submodule pin bumped, distribute artifact shipped, new operator-blocked open question surfaced).
6. **Run `scripts/verify-all-constitution-rules.sh`** before any release-tagging or major-state-change attempt. The sweep wrapper is in fully STRICT mode; a non-40/40 result is a release blocker.
7. **Re-generate the coverage ledger** (`scripts/generate-coverage-ledger.sh`) in the same commit as any new feature module or any module-deletion; the STRICT-mode gate rejects stale-ledger commits.

8. **The autonomous loop ends** when the next forward step requires operator-environment access (real device, real keystore secrets, Firebase token, ssh credentials) OR operator decision-making (open questions, brainstorming next phase scope, tagging, choosing a UI direction). At that point, summarize state + ask the operator the specific next-step question.

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9. Cross-references

- **Plan docs (this cycle):**
  - `docs/plans/2026-05-15-constitution-compliance.md` — master plan
  - `docs/plans/2026-05-16-helixqa-integration-design.md` — Option 1 wiring (DONE)
  - `docs/plans/2026-05-16-helixqa-go-package-linking-design.md` — Option 2 design (DESIGN-ONLY; operator-blocked)
  - `docs/plans/2026-05-16-snake_case-migration.md` — Phase 6 plan (PLAN-ONLY; operator-blocked)
- **Gates inventory:** `docs/helix-constitution-gates.md`
- **Coverage ledger:** `docs/coverage-ledger.yaml` + `docs/coverage-ledger.waivers.yaml`
- **Constitution source-of-truth:** `constitution/` submodule (HelixConstitution at `464ada14`)
- **Sweep wrapper:** `scripts/verify-all-constitution-rules.sh`
- **Sweep attestations:** `.lava-ci-evidence/verify-all/<UTC-timestamp>.json`
- **HelixQA audit:** `docs/helixqa-script-audit.md`
- **HelixQA wrapper:** `scripts/run-helixqa-challenges.sh`
- **HelixQA hermetic test:** `tests/check-constitution/test_helixqa_wiring.sh`